



Always-on display on iOS 16

Apple is expected to show off the much awaited always-on lock screen for iOS 16.
<https://digit.in/jun22-50>



Frank Klein new COO of Rivian

The automaker company hired a new COO Frank Klein who joined on June 1.
<https://digit.in/jun22-51>

Barcodes

Explore the origins of the ubiquitous barcodes

Paarth Pareek | feedback@digit.in

If you've ever been to a supermarket, you've seen barcodes being scanned around with specific scanners which have eased the lives of every supermarket clerk. However, getting to the barcodes that we see today, took longer than you might expect. Unlike what most people think, barcodes have been in development for 70 years!

THE FIRST BARCODE

The first barcode was made in 1948 by students of Drexel University. Their sole objective was better inventory management and check-outs for various supermarket chains and to make a change that would become universal.

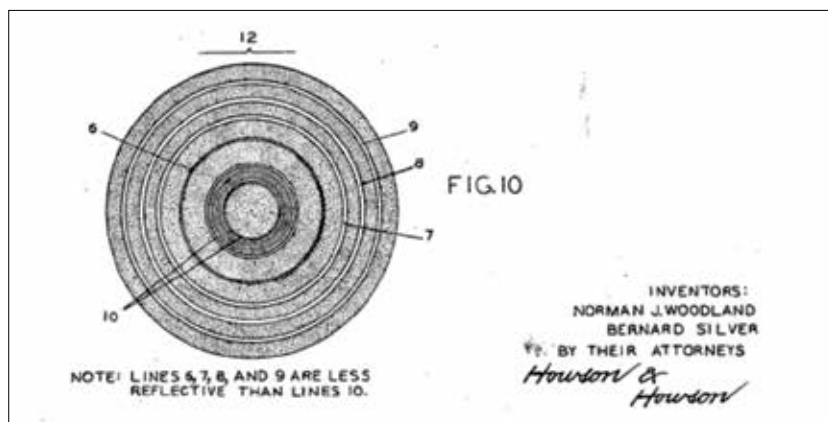
Though it is considered a success today, the invention wasn't received all that well back in the day because of the lack of technological development. Back then, barcode scanners were exorbitantly priced and weren't nearly as mainstream as they are now.

THE FIRST IMPLEMENTATION

While the invention of the barcode dated back to 1948 and the patent was received by its inventors in 1952, it wasn't until 1960 that the barcode saw its first ever usage in the practical world. The Association of American Railroads had been having problems in tracking their rail cars and wanted to streamline the process to make

tracking quicker and easier. This was when their attention was brought towards the barcode patent and how they could do something similar. It was David J. Collins who designed the system. He was an undergraduate at MIT and was working for the Pennsylvania Railroad.

Once the design was done, the Association of American Railroads sponsored the project which was



The Patent image for the first barcode by Norman Woodland and Bernard Silver

built by a company called Sylvania. The system was named KarTrak ACI (Automatic Car Identification).

The system used a pattern of 3M Scotch lite blue and orange stripes which encoded ownership details along with a unique car number



KarTrak Code

that would be specific to a particular rail car.

Even though the barcode itself was simple to make, using only 3M Scotchlite strips, the scanner was the real issue. It was about the size of a modern refrigerator and would activate as a train approached. It would shine a 500-watt light on the carriages while a sensor interpreted the reflections being given out by the Scotchlite strips.

The data received by the sensor would be printed on a teletype machine, paper, or magnetic tapes, and in some special cases, the data could directly be recorded to a computer.

It wasn't long until KarTrak became the norm for all railroads that existed. KarTrak became the US Standard in 1968, and by 1975, over 90 percent of the US Railcars carried a KarTrak label. However, KarTrak's downfall came even more swiftly than its success. In 1978, it was abandoned because of high maintenance costs of the constantly wearing out labels, poor read accuracy, highly expensive computers which were needed in running the system, and of course, the bankruptcies faced by the Railroads.

LASER IT DOWN!

While KarTrak was drowning in its success, Collins decided to quit Sylvania and co-founded Computer Identities Corporation, the company which would revolutionise barcodes